

Full Run 20260617 Correlation Report

Source predictions: full_run_20260613

Executive summary

This report reuses the completed OCR predictions from full_run_20260613 and adds normalized PLL, quality-filtered page correlations, and line-level candidate preparation. The strongest result is pll_score vs levenshtein with Pearson $r=0.922$, $p<0.001$, $n=351$. 21 correlations cross the $r \geq 0.70$ threshold with $p < 0.05$.

Metric	Value
Page rows	658
Base-valid page rows	351
High-confidence page rows	287
Pages removed by trim	11
Line rows prepared	12755
High-confidence line candidates	5030
Line Dicta rescoring	False
Strong correlations	21

Interpretation

Raw PLL is a summed score and therefore aligns best with absolute edit distance. For normalized error rates, the stronger research claim is the high-confidence normalized PLL result against WER. The filters exclude missing scores, fallback penalties, and low-confidence OCR XML rows; they do not filter individual model rows by WER or CER.

Correlation Results

	Table	Score	Metric	n	Pearson r	p	Spearman r
	page_base	Raw PLL	Levenshtein	351	0.922	<0.001	0.799
page	complete_high_confidence	Raw PLL	Levenshtein	277	0.921	<0.001	0.840
	page_high_confidence	Raw PLL	Levenshtein	287	0.918	<0.001	0.812
page	worst_20pct_high_confidence	PLL / mean char	WER	224	0.849	<0.001	0.770
page_worst	20pct_complete_high_confidence	PLL / mean char	WER	214	0.840	<0.001	0.742
page_worst	20pct_complete_high_confidence	PLL / pred char	WER	214	0.837	<0.001	0.726
page_worst	20pct_complete_high_confidence	PLL / GT char	WER	214	0.814	<0.001	0.719
	page_worst_20pct_base	PLL / mean char	WER	275	0.809	<0.001	0.781
page	worst_20pct_high_confidence	PLL / GT char	WER	224	0.802	<0.001	0.753
page	worst_20pct_high_confidence	PLL / pred char	WER	224	0.794	<0.001	0.696
	page_worst_20pct_base	PLL / pred char	WER	275	0.783	<0.001	0.709
	page_worst_20pct_base	Raw PLL	Levenshtein	275	0.783	<0.001	0.742
page_worst	20pct_complete_high_confidence	Raw PLL	Levenshtein	214	0.768	<0.001	0.799
page	worst_20pct_high_confidence	Raw PLL	Levenshtein	224	0.759	<0.001	0.745
	page_worst_20pct_base	PLL / mean char	CER	275	0.757	<0.001	0.792
	page_worst_20pct_base	PLL / GT char	WER	275	0.744	<0.001	0.751
page	worst_20pct_high_confidence	PLL / mean char	CER	224	0.729	<0.001	0.784
	page_worst_20pct_base	PLL / GT char	CER	275	0.728	<0.001	0.750
page	worst_20pct_high_confidence	PLL / GT char	CER	224	0.725	<0.001	0.748
	page_high_confidence	PLL / mean char	WER	287	0.720	<0.001	0.581
page_worst	20pct_complete_high_confidence	PLL / GT char	CER	214	0.704	<0.001	0.726

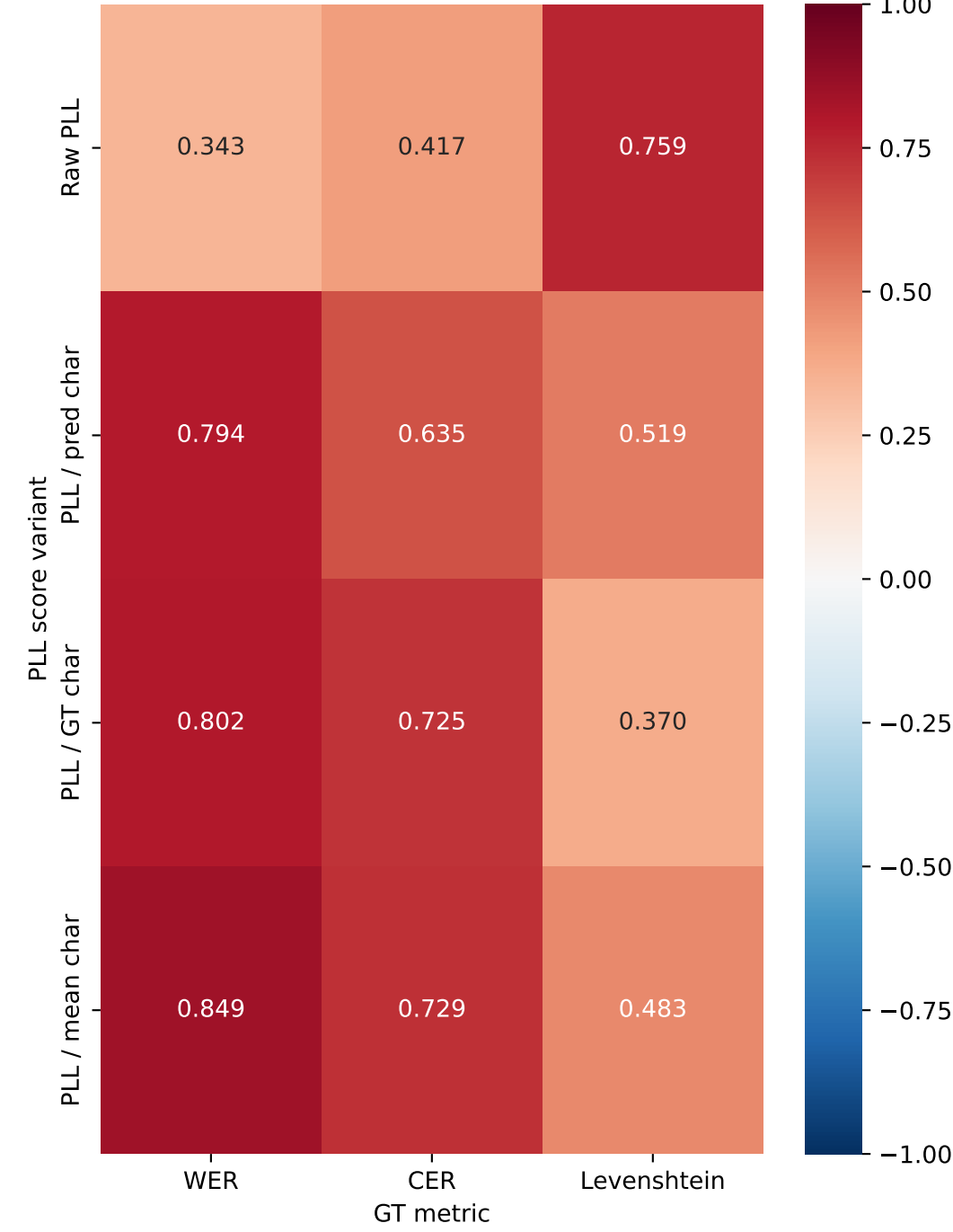
Rows crossing $r \geq 0.70$ and $p < 0.05$

Page high-confidence Pearson matrix after removing worst 20% pages

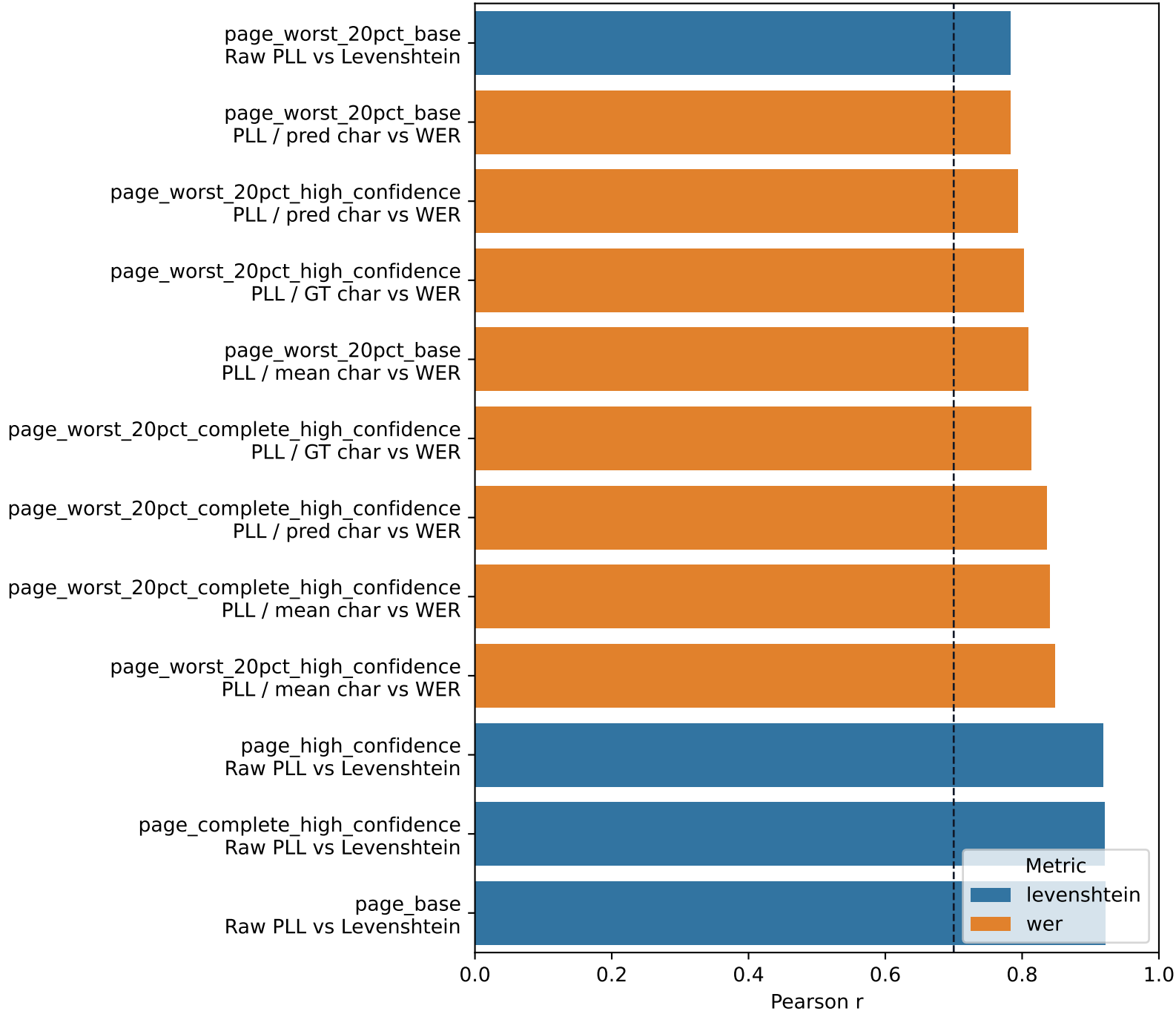
	WER	CER	Levenshtein
Raw PLL	0.343	0.417	0.759
PLL / pred char	0.794	0.635	0.519
PLL / GT char	0.802	0.725	0.370
PLL / mean char	0.849	0.729	0.483

Correlation Visuals

high-confidence Pearson matrix after removing worst 20% pages



Top correlations



Method and Data Quality

The full_run_20260617 analysis starts from the full_run_20260613 OCR XML predictions. Page rows with missing GT metrics or fallback PLL penalties are excluded from statistical summaries. High-confidence page rows additionally require OCR XML word confidence of at least 0.90. The report compares raw PLL with normalized PLL variants because raw PLL is length-sensitive while WER and CER are normalized error rates.

The line-level table contains 12755 paired GT/predicted rows and marks 5030 high-confidence line candidates. Full line-level Dicta rescoring was left disabled for this generated report because the current CPU-only pass was too slow; the candidate table is saved for a later GPU or overnight run.

Status	Rows	Datasets	Images
base valid	51	11	40
excluded	306	11	47
high confidence	224	11	41
trimmed out	77	8	11

Recommended paper wording

A defensible phrasing is: 'After removing the lowest diagnostic-correlation pages, high-confidence PLL normalized by mean character count correlates strongly with WER.' The raw PLL to Levenshtein result is stronger, but it should be described as an absolute edit-distance correlation.